

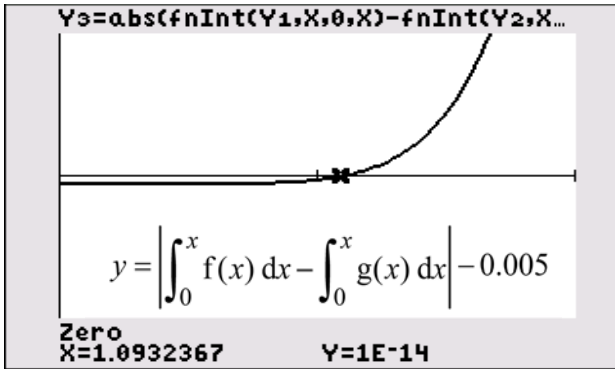
Innova Junior College
H2 Mathematics
2018 Prelim Exam Paper 1 Solution

Q1	Suggested Solution
	$\frac{dy}{dx} = \sqrt{y^2 + e^{2x}} \Rightarrow \left(\frac{dy}{dx}\right)^2 = y^2 + e^{2x} \text{ --- (1)}$ Differentiating (1) with respect to x. $2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} = 2y\frac{dy}{dx} + 2e^{2x} \text{ --- (2)}$ When $x = 0$, $y = 1$ (given) $\frac{dy}{dx} = \sqrt{1 + e^0} = \sqrt{2}$ From (2): $2\sqrt{2}\frac{d^2y}{dx^2} = 2(1)(\sqrt{2}) + 2e^0 \Rightarrow \frac{d^2y}{dx^2} = \frac{\sqrt{2} + 1}{\sqrt{2}}$ Therefore the Maclaurin series for y is $y = 1 + x\sqrt{2} + \left(\frac{1 + \sqrt{2}}{2\sqrt{2}}\right)x^2 + \dots$
Q2	Suggested Solution
	$\int_0^{\pi} x^2 \sin(nx) dx$ $= \left[-x^2 \frac{\cos(nx)}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} 2x \cos(nx) dx \text{ --- (1)}$ $= -\frac{\pi^2}{n} \cos(n\pi) + \frac{1}{n} \left\{ \left[(2x) \frac{\sin(nx)}{n} \right]_0^{\pi} - \int_0^{\pi} (2) \frac{\sin(nx)}{n} dx \right\}$ $= -\frac{\pi^2}{n} - \frac{2}{n^2} \int_0^{\pi} \sin(nx) dx \text{ --- (2)}$ $= -\frac{\pi^2}{n} - \frac{2}{n^2} \left[\frac{-\cos(nx)}{n} \right]_0^{\pi}$ $= -\frac{\pi^2}{n} + \frac{2}{n^3} [\cos(n\pi) - \cos 0]$ $= -\frac{\pi^2}{n}$ $\therefore k = -1$
Q3	Suggested Solution
(i)	Since p and q are non-parallel vectors, for some α and β , the sum of $\alpha\mathbf{p}$ and $\beta\mathbf{q}$ is r by law of parallelogram for vector addition.

(ii)	Using ratio theorem, $\frac{\vec{OX}}{\vec{OR}} = \frac{\vec{OX}}{\vec{OR}} = \frac{1}{3}\vec{p} + \frac{2}{3}\vec{r} = \frac{1}{3}\vec{p} + \frac{2}{3}\alpha\vec{p} + \frac{2}{3}\beta\vec{q}$ Area of triangle OPX $= \frac{1}{2} \vec{p} \times \vec{x} $ $= \frac{1}{2} \vec{p} \times \left(\frac{1}{3}\vec{p} + \frac{2}{3}\alpha\vec{p} + \frac{2}{3}\beta\vec{q}\right) $ $= \frac{1}{3}\beta \vec{p} \times \vec{q} $ Area of triangle OQR $= \frac{1}{2} \vec{q} \times \vec{r} $ $= \frac{1}{2} \vec{q} \times (\alpha\vec{p} + \beta\vec{q}) $ $= \frac{1}{2}\alpha \vec{p} \times \vec{q} $ Since the area of the triangles are the same, $\frac{1}{3}\beta \vec{p} \times \vec{q} = \frac{1}{2}\alpha \vec{p} \times \vec{q} $ $\frac{\alpha}{\beta} = \frac{2}{3}$ The ratio required is 2:3.
Q4	Suggested Solution
(a)	$x^2(x-5) \geq (x-5)(2kx-k^2)$ $(x-5)\left[x^2 - (2kx-k^2)\right] \geq 0$ $(x-5)(x^2 - 2kx + k^2) \geq 0$ $(x-5)(x-k)^2 \geq 0$ $\begin{array}{c} - \qquad \qquad - \qquad \qquad + \\ \hline \qquad \qquad \qquad \qquad \\ \qquad \qquad k \qquad \qquad 5 \end{array}$ $x = k$ or $x \geq 5$
(b)	$f(x) = a + bx - x^2$; $g(x) = x - c$ where $2 < c < 4$ $2 < c \Rightarrow g(2) = 2 - c = c - 2$ $c < 4 \Rightarrow g(5) = 5 - c = 5 - c$ $f(2) = g(2) \Rightarrow a + 2b - 4 = c - 2$ $a + 2b - c = 2$ L L (1)

	$f(5) = g(5) \Rightarrow a + 5b - 25 = 5 - c$ $a + 5b + c = 30 \text{ L L (2)}$ $f(4) - g(4) = \frac{8}{3} \Rightarrow a + 4b - 16 - (4 - c) = \frac{8}{3}$ $a + 4b + c = \frac{68}{3} \text{ L L (3)}$ $a = -\frac{29}{3}; \quad b = \frac{22}{3}; \quad c = 3$ Area bounded $\int_2^3 -\frac{29}{3} + \frac{22}{3}x - x^2 + (x - 3) dx$ $= \left[-\frac{29}{3}x + \frac{11}{3}x^2 - \frac{x^3}{3} + \frac{1}{2}x^2 - 3x \right]_2^3$ $= \left(-29 + 33 - 9 + \frac{9}{2} - 9 \right) - \left(-\frac{58}{3} + \frac{44}{3} - \frac{8}{3} + 2 - 6 \right)$ $= \frac{11}{6}$
Q5	Suggested Solution
(a)	$\sum_{r=1}^N \frac{1}{(r-1)! + r!} = \sum_{r=1}^N (u_r - u_{r+1})$ $= \cancel{u_1} - \cancel{u_2}$ $+ \cancel{u_2} - \cancel{u_3}$ $\quad \quad \quad \text{M}$ $+ \cancel{u_{N-1}} - \cancel{u_N}$ $+ \cancel{u_N} - u_{N+1}$ $= u_1 - u_{N+1}$ $= 1 - \frac{1}{(N+1)!}$ $\sum_{r=4}^N \frac{1}{(r-1)! + r!} = \sum_{r=1}^N \frac{1}{(r-1)! + r!} - \sum_{r=1}^3 \frac{1}{(r-1)! + r!}$ $= \left(1 - \frac{1}{(N+1)!} \right) - \left(1 - \frac{1}{4!} \right)$ $= \frac{1}{24} - \frac{1}{(N+1)!}$ $< \frac{1}{24} \quad \left(\text{since } \frac{1}{(N+1)!} > 0 \right)$

(b)	Let $a_n = \frac{(-1)^n \pi^{2n}}{(2n)!}$ $\frac{a_{n+1}}{a_n} = \frac{(-1)^{n+1} \pi^{2n+2}}{(2n+2)!} \times \frac{(2n)!}{(-1)^n \pi^{2n}}$ $= -\frac{\pi^2}{(2n+1)(2n+2)}$ $\lim_{n \rightarrow \infty} \left \frac{a_{n+1}}{a_n} \right = \lim_{n \rightarrow \infty} \frac{\pi^2}{(2n+1)(2n+2)} = 0 < 1$ Hence by ratio test, $\sum_{r=0}^{\infty} \frac{(-1)^r \pi^{2r}}{(2r)!}$ converges. $\sum_{r=0}^{\infty} \frac{(-1)^r \pi^{2r}}{(2r)!} = \cos \pi = -1$
Q6	Suggested Solution
(i)	$\int_0^k f(x) dx = \int_0^k x(2+x^2)^{-1/2} dx$ $= \frac{1}{2} \int_0^k 2x(2+x^2)^{-1/2} dx$ $= \frac{1}{2} \left[\frac{(2+x^2)^{1/2}}{1/2} \right]_0^k$ $= \sqrt{2+k^2} - \sqrt{2}$
(ii)	$f(x) = x(2+x^2)^{-1/2}$ $= x \left[2 \left(1 + \frac{x^2}{2} \right) \right]^{-1/2}$ $= \frac{1}{\sqrt{2}} x \left(1 - \frac{1}{2} \cdot \frac{x^2}{2} + \frac{-\frac{1}{2}(-\frac{3}{2})}{2!} \left(\frac{x^2}{2} \right)^2 + \dots \right)$ $= \frac{1}{\sqrt{2}} x \left(1 - \frac{1}{4} x^2 + \frac{3}{32} x^4 + \dots \right)$ $g(x) = \frac{1}{\sqrt{2}} x - \frac{1}{4\sqrt{2}} x^3 + \frac{3}{32\sqrt{2}} x^5$

	The expansion is valid when $\left \frac{x^2}{2}\right < 1$, $\Rightarrow x^2 < 2$ $ x < \sqrt{2}$ $\{x: x \in \mathbb{R}, -\sqrt{2} < x < \sqrt{2}\}$
(iii)	$\left \int_0^k f(x) dx - \int_0^k g(x) dx \right < 0.005$ $\left \int_0^k f(x) dx - \int_0^k g(x) dx \right - 0.005 < 0$  From GC, $0 < k < 1.093$ (4 s.f.)
Q7	Suggested Solution
(i)	1st term = T & Common difference = 1.5 Total time taken to complete the distance of 4 km (i.e. 80 laps) $= \frac{80}{2} [2T + (80-1)1.5] = 80T + 4740$ To complete within the required time interval, $\left(2\frac{1}{3}\right)(60)(60) \leq 80T + 4740 \leq \left(2\frac{5}{6}\right)(60)(60)$ $8400 \leq 80T + 4740 \leq 10200$ $45.75 \leq T \leq 68.25$ Set of values of T is $\{T \in \mathbb{R} : 45.75 \leq T \leq 68.25\}$
(ii)	1st term = t & Common ratio = 1.015 Total time taken to complete the distance of 4 km (i.e. 80 laps) $= \frac{t(1.015^{80} - 1)}{1.015 - 1} = \frac{200t}{3} (1.015^{80} - 1)$ To complete within the required time interval, $8400 \leq \frac{200t}{3} (1.015^{80} - 1) \leq 10200$ $55.0059 \leq t \leq 66.7928$ $55.01 \leq t \leq 66.79 \text{ (2 dec pl)}$

	Set of values of t is $\{t \in \mathbb{R} : 55.01 \leq t \leq 66.79\}$ (to 2d.p.)
(iii)	Completing a distance of 1.8 km is equivalent to swimming 36 laps For swimmer A: $\frac{36}{2}[2T + (36-1)1.5] = (50)(60) \Rightarrow T = 57.083333$ $\therefore$ time taken to swim the 80th lap $= 57.083333 + (80-1)(1.5) = 175.58333$ For swimmer B: $\frac{t(1.015^{36} - 1)}{1.015 - 1} = (50)(60) \Rightarrow t = 63.457186$ $\therefore$ time taken to swim the 80th lap $= 63.457186(1.015)^{80-1} = 205.73024$ Swimmer A is faster in his 80th lap.
Q8	Suggested Solution
(a)	$(3+i)z + 3w = -5i$ --- (1) $(i-2)z - 6iw = 1-3i$ --- (2) $(1) \times 2i : 2i(3+i)z + 6iw = -5i(2i)$ --- (3) $(2) + (3):$ $(i-2)z + 2i(3+i)z = 1-3i-5i(2i)$ $z(i-2+6i-2) = 1-3i+10$ $z = \frac{11-3i}{-4+7i} \times \frac{-4-7i}{-4-7i}$ $= \frac{-44-77i+12i-21}{16+49}$ $= \frac{-65-65i}{65}$ $= -1-i$ Substitute $z = -1-i$ into (1): $(3+i)(-1-i) + 3w = -5i$ $3w = -5i - (-3-3i-i+1)$ $w = \frac{2-i}{3}$

(b)(i)	Since the coefficients of the equation are real, $-\frac{1}{2}(1-i)$ is another root of the equation. Quadratic factor $= \left(\omega + \frac{1}{2} + \frac{1}{2}i\right)\left(\omega + \frac{1}{2} - \frac{1}{2}i\right)$ $= \left(\omega + \frac{1}{2}\right)^2 - \left(\frac{1}{2}i\right)^2$ $= \omega^2 + \omega + \frac{1}{4} + \frac{1}{4}$ $= \omega^2 + \omega + \frac{1}{2}$ By inspection, $k\omega^4 - 2\omega^3 + 5\omega^2 + 6\omega + 4 = \left(\omega^2 + \omega + \frac{1}{2}\right)(k\omega^2 + p\omega + 8)$ Comparing ω: $6 = 8 + \frac{1}{2}p \Rightarrow p = -4$ Comparing ω^3: $-2 = p + k \Rightarrow k = -2 + 4 = 2$ Solving $2\omega^2 - 4\omega + 8 = 0$: $\omega = \frac{4 \pm \sqrt{16 - 4(2)(8)}}{2(2)} = \frac{4 \pm 4i\sqrt{3}}{4} = 1 \pm i\sqrt{3}$
(b)(ii)	Note that $\omega_3 = 1 + i\sqrt{3}$ and $\omega_4 = -\frac{1}{2}(1-i)$. $\left \frac{\omega_3}{\omega_4}\right = \frac{ \omega_3 }{ \omega_4 } = \frac{2}{\frac{\sqrt{2}}{2}} = \frac{4}{\sqrt{2}}$ $\arg\left(\frac{\omega_3}{\omega_4}\right) = \arg \omega_3 - \arg \omega_4 = \frac{\pi}{3} - \frac{3\pi}{4} = -\frac{5\pi}{12}$ $\frac{\omega_3}{\omega_4} = \frac{4}{\sqrt{2}} \left[\cos\left(-\frac{5\pi}{12}\right) + i \sin\left(-\frac{5\pi}{12}\right) \right]$
9	Suggested Solution
(i)	$\frac{dy}{dx} = \frac{(x-4)(4x+k) - (2x^2 + kx + 8)}{(x-4)^2}$ $= \frac{4x^2 + (k-16)x - 4k - 2x^2 - kx - 8}{(x-4)^2}$ $= \frac{2x^2 - 16x - 4k - 8}{(x-4)^2}$ When $x = 1$, $y = \frac{10+k}{-3}$, and $\frac{dy}{dx} = \frac{2 - 16 - 4k - 8}{(-3)^2} = \frac{-22 - 4k}{9}$

Equation of tangent to C at $x = 1$:

$$y - \frac{10+k}{-3} = \frac{-22-4k}{9}(x-1)$$

$$9y + 30 + 3k = (-22 - 4k)x + 22 + 4k$$

$$9y + 2(11 + 2k)x = k - 8$$

$$\text{Or } y = -\frac{2(11+2k)}{9}x + \frac{k-8}{9}$$

(ii)

For stationary points,

$$\frac{dy}{dx} = 0$$

$$\frac{2x^2 - 16x - 4k - 8}{(x-4)^2} = 0$$

$$x^2 - 8x - 2k - 4 = 0$$

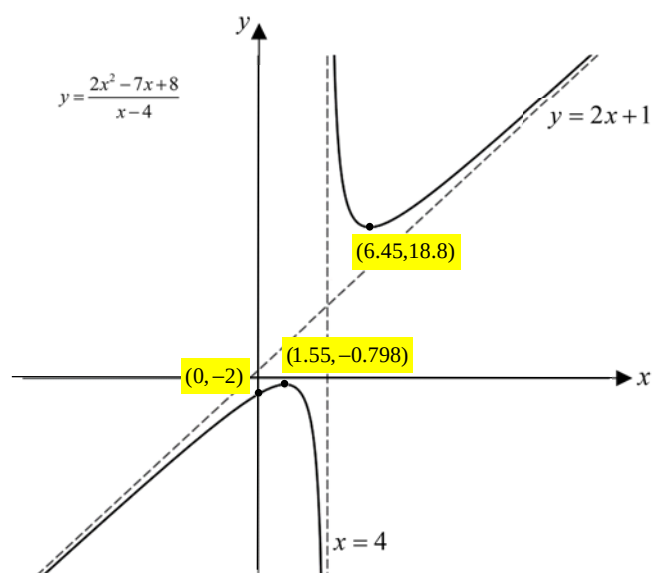
For more than 1 stationary point, this equation must have real and distinct roots,

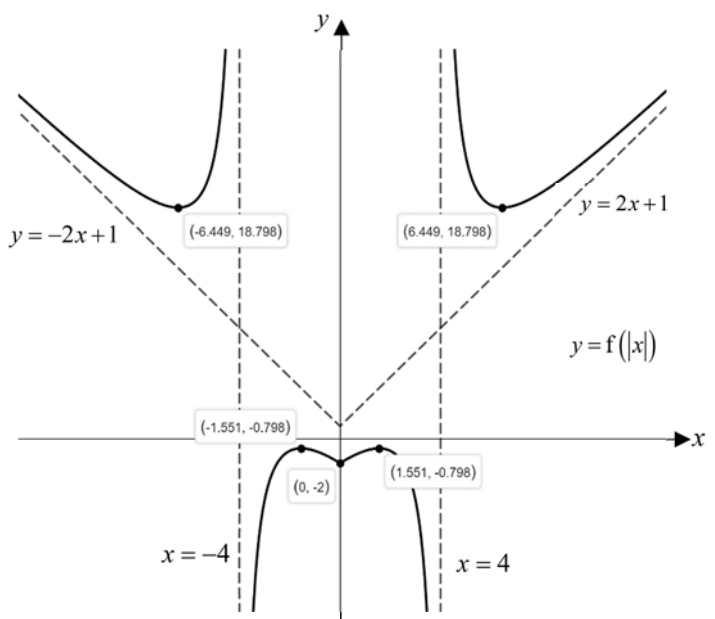
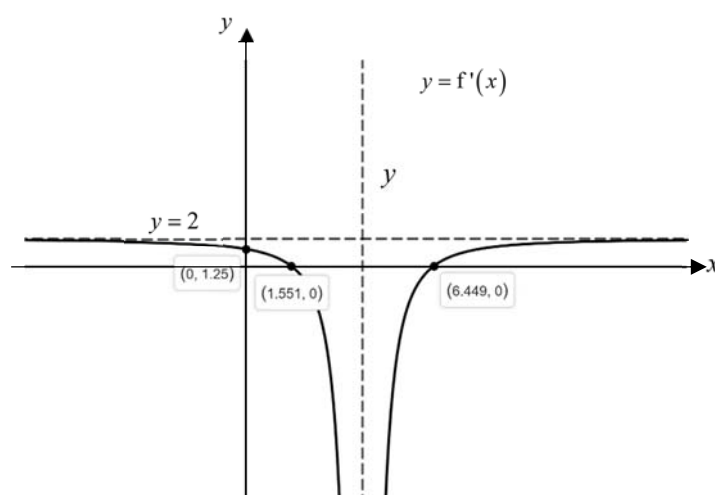
$$(-8)^2 - 4(-2k - 4) > 0$$

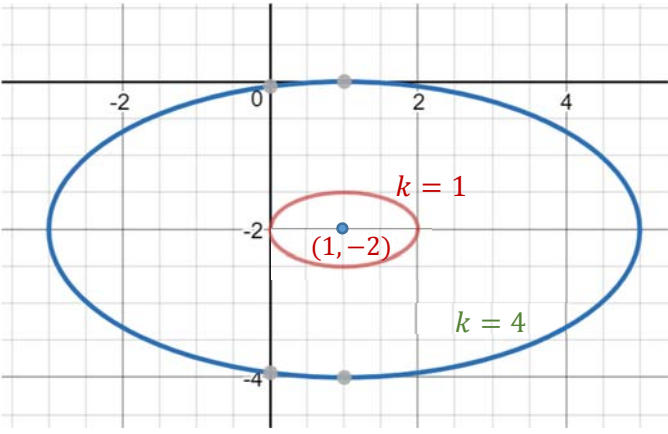
$$64 + 8k + 16 > 0$$

$$k > -10$$

(iii)



(iv) (a)	
(b)	
Q10 Suggested Solution	
(i)	$x = 1 + k \cos \theta \Rightarrow \cos \theta = \frac{(x-1)}{k}$ $y = -2 + \frac{1}{2} k \sin \theta \Rightarrow \sin \theta = \frac{2(y+2)}{k}$ $\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \frac{(x-1)^2}{k^2} + \frac{4(y+2)^2}{k^2} = 1$ Cartesian eqn of C_k is $(x-1)^2 + 4(y+2)^2 = k^2$ Applying implicit differentiation, $2(x-1) + 8(y+2) \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{(x-1)}{4(y+2)} = \frac{1-x}{4(y+2)}$

(ii)	
(iii)	The path of the stream on the map is perpendicular to the ellipses implies that $(\text{gradient of the path}) \times (\text{grad of ellipse}) = -1$. Since gradient of ellipse $= -\frac{(x-1)}{4(y+2)}$, We have $\frac{dy}{dx} \left[-\frac{(x-1)}{4(y+2)} \right] = -1$ Thus $\frac{1}{y+2} \times \frac{dy}{dx} = \frac{4}{x-1}$. Using the given result, $\int \left(\frac{1}{y+2} \right) dy = \int \frac{4}{x-1} dx$ $\ln y+2 = 4 \ln x-1 + C$ $ y+2 = e^{4 \ln x-1 + C}$ $= (e^{\ln x-1 ^4}) (e^C)$ $y+2 = \pm (e^C) (x-1)^4$ $y = A(x-1)^4 - 2 \quad \text{where } A = \pm e^C$
(iv)	The path passes through the point $(-1, -1)$, $-1 = A(-1-1)^4 - 2$ $A = \frac{1}{16}$ Eqn of the path is $y = \frac{1}{16}(x-1)^4 - 2$

Q11	Suggested Solution
(i)	$\vec{AB} = \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \\ 5 \end{pmatrix} = 6 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \Rightarrow \vec{AB} = 6\sqrt{2}$ $\vec{AC} = \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \\ 5 \end{pmatrix} = -6 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \vec{AC} = 6\sqrt{2}$ $\vec{BC} = \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix} = -6 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \vec{BC} = 6\sqrt{2}$ Since $AB = BC = CA$, triangle ABC is an equilateral triangle.
(ii)	$\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
(iii)	Since π_1 is perpendicular to $\vec{AB}$, the normal vector of π_1 is $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$. By symmetry, π_1 will pass through C. $\pi_1 : r \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = -1$ Cartesian equation of π_1 is: $y - z = -1$
(iv)	Since π_2 is perpendicular to $\vec{BC}$, the normal vector of π_2 is $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. By symmetry, π_2 will pass through A. $\pi_2 : r \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 3$ Cartesian equation of π_2 is: $x + y = 3$ Using GC, the equation of the line of intersection of the two planes is $l : r = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \text{ where } \lambda \in \mathbb{R}.$

(v)	Note that point G lies on the line l found in part (iv). Since G lies on l, $\vec{OG} = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$ for some $\lambda \in \mathbb{R}$; $\vec{AG} = \begin{pmatrix} -1-\lambda \\ 1+\lambda \\ -5+\lambda \end{pmatrix}$ & $\vec{DG} = \begin{pmatrix} -3-\lambda \\ 3+\lambda \\ \lambda+3 \end{pmatrix}$ Given that $\vec{DG} = \vec{AG}$, $2(1+\lambda)^2 + (\lambda-5)^2 = 3(\lambda+3)^2$ $2(\lambda^2 + 2\lambda + 1) + (\lambda^2 - 10\lambda + 25) = 3(\lambda^2 + 6\lambda + 9)$ $4\lambda + 2 - 10\lambda + 25 = 18\lambda + 27$ $\lambda = 0$ $\vec{OG} = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}$ Thus, coordinates of G are $(4, -1, 0)$.
(vi)	$\cos \angle AGD = \frac{\vec{DG} \cdot \vec{AG}}{ \vec{DG} \vec{AG} } = \frac{3 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ -5 \end{pmatrix}}{(3\sqrt{3})(\sqrt{27})} = -\frac{1}{3}$ $\angle AGD = 109.5^\circ$
	Alternative: Let angle AGX be α. $\cos \alpha = \frac{1}{3} \Rightarrow \alpha = 70.52^\circ \Rightarrow \theta = 180^\circ - \alpha = 109.5^\circ$